

7. (a) The table shows the first four members of the alkane family.

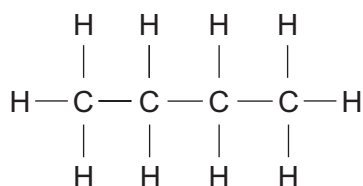
Alkanes
methane, CH ₄
ethane, C ₂ H ₆
propane, C ₃ H ₈
butane, C ₄ H ₁₀

Give the **general** formula for the alkane family.

[1]

.....

- (b) C₄H₁₀ has two isomers. The diagram below shows the structure of one of the isomers, butane.



Draw and name the other isomer.

[2]

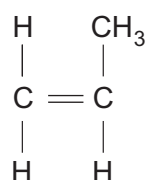
Isomer structure

Name of isomer



(c) The flow diagram shows two reactions of propene.

Draw the structural formula for compound **A** and the repeating unit in polymer **B**. [2]



bromine, Br₂

A

polymerisation

B

(d) A student was given a colourless liquid. He suspects the liquid is ethanol.

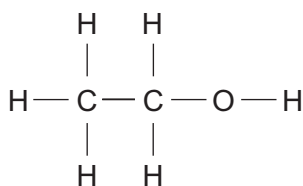
Describe a chemical test that he could carry out to positively identify the liquid as ethanol. Include the observation he would make. [2]

Test

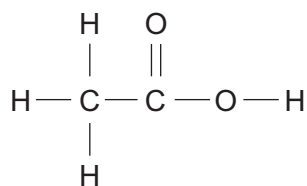
Observation



(e) Organic compounds can also be identified using infrared spectroscopy.



ethanol



ethanoic acid

Bond	Wavenumber (cm^{-1})
$\text{C}=\text{O}$	1650 to 1750
$\text{C}-\text{H}$	2800 to 3100
$\text{O}-\text{H}$	2500 to 3550

The infrared spectra of ethanol, ethanoic acid and one other compound are shown on the opposite page.

Use the information in the table to identify the spectra belonging to ethanol and ethanoic acid. [2]

Ethanol

Ethanoic acid



